WORKED SOLUTION

STEP I 2018, Question 8

Pure • Coupled Differential Equations

> **STEP I 2018, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/18-s1-q8

The key idea

The two derivative rules are built to cancel against each other: whenever s' appears it brings in c^2, and whenever c' appears it brings in $-\mathrm{s}^2$. So the natural move throughout is to differentiate products and quotients and watch the terms combine, then use the identity from (i) to turn everything into a single function. For the integrals, read each integrand as "something times s^2" or "something times c^2", because those are exactly the derivatives we know. The u-integrals in the last two parts are the same integrals in disguise.

Solution

Throughout, write s and c for $\mathrm{s}(x)$ and $\mathrm{c}(x)$ when there is no ambiguity.

Part (i)

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{s}^3+\mathrm{c}^3\right) = 3\mathrm{s}^2\,\mathrm{s}' + 3\mathrm{c}^2\,\mathrm{c}' = 3\mathrm{s}^2\mathrm{c}^2 - 3\mathrm{c}^2\mathrm{s}^2 = 0,$$

so $\mathrm{s}^3+\mathrm{c}^3$ is constant. Its value at $x=0$ is $0^3+1^3=1$, hence

$$\mathrm{s}(x)^3+\mathrm{c}(x)^3 = 1 \quad \text{for all } x,$$

as required. This identity is the engine of the rest of the question: it lets us swap s^3 for $1-\mathrm{c}^3$ and vice versa.

Part (ii)

By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{sc}) = \mathrm{s}'\mathrm{c} + \mathrm{sc}' = \mathrm{c}^2\cdot\mathrm{c} + \mathrm{s}\cdot(-\mathrm{s}^2) = \mathrm{c}^3 - \mathrm{s}^3.$$

Using (i), $\mathrm{s}^3 = 1 - \mathrm{c}^3$, so this equals $\mathrm{c}^3 - (1 - \mathrm{c}^3) = 2\mathrm{c}^3 - 1$, which is the result we wanted.

For the quotient (wherever $\mathrm{c}(x) \neq 0$), the quotient rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{s}}{\mathrm{c}}\right) = \frac{\mathrm{s}'\mathrm{c} - \mathrm{s}\mathrm{c}'}{\mathrm{c}^2} = \frac{\mathrm{c}^3 + \mathrm{s}^3}{\mathrm{c}^2} = \frac{1}{\mathrm{c}(x)^2},$$

again by (i). Notice the pleasing parallel with $\frac{\mathrm{d}}{\mathrm{d}x}\tan x = \frac{1}{\cos^2 x}$.

Part (iii)

Since $\mathrm{c}' = -\mathrm{s}^2$, the first integral is immediate:

$$\int \mathrm{s}(x)^2 \,\mathrm{d}x = -\mathrm{c}(x) + C.$$

Ask yourself How can s^5 be written so that the known derivative $\mathrm{c}' = -\mathrm{s}^2$ appears?

Split off a factor s^2 and use (i) on what is left: $\mathrm{s}^5 = \mathrm{s}^3 \cdot \mathrm{s}^2 = (1 - \mathrm{c}^3)\,\mathrm{s}^2 = -(1 - \mathrm{c}^3)\,\mathrm{c}'$. This is of the form $g(\mathrm{c})\,\mathrm{c}'$, which integrates by the reverse chain rule:

$$\int \mathrm{s}(x)^5 \,\mathrm{d}x = -\int (1 - \mathrm{c}^3)\,\mathrm{c}' \,\mathrm{d}x = -\mathrm{c}(x) + \tfrac{1}{4}\mathrm{c}(x)^4 + C.$$

(Differentiating back: $-\mathrm{c}' + \mathrm{c}^3\mathrm{c}' = -\mathrm{c}'(1 - \mathrm{c}^3) = \mathrm{s}^2 \cdot \mathrm{s}^3$, as it should be.)

Part (iv)

Put $u = \mathrm{s}(x)$, so $x = \mathrm{s}^{-1}(u)$ and $\mathrm{d}u = \mathrm{s}'(x)\,\mathrm{d}x = \mathrm{c}(x)^2\,\mathrm{d}x$. From (i), $\mathrm{c}^3 = 1 - u^3$, so $\mathrm{c} = (1 - u^3)^{1/3}$ (the real cube root, which is valid whatever the sign of c), and therefore

$$(1 - u^3)^{2/3} = \mathrm{c}(x)^2.$$

Hence

$$\int \frac{1}{(1 - u^3)^{2/3}}\,\mathrm{d}u = \int \frac{1}{\mathrm{c}^2}\,\mathrm{c}^2\,\mathrm{d}x = \int 1\,\mathrm{d}x = x + C = \mathrm{s}^{-1}(u) + C,$$

which is what we had to show.

Part (v)

Use the same substitution, with $(1 - u^3)^{1/3} = \mathrm{c}$ and $\mathrm{d}u = \mathrm{c}^2\,\mathrm{d}x$.

For the first integral, $(1 - u^3)^{4/3} = \mathrm{c}^4$, so, by the quotient result in (ii),

$$\int \frac{\mathrm{d}u}{(1 - u^3)^{4/3}} = \int \frac{\mathrm{c}^2}{\mathrm{c}^4}\,\mathrm{d}x = \int \frac{\mathrm{d}x}{\mathrm{c}^2} = \frac{\mathrm{s}(x)}{\mathrm{c}(x)} + C = \frac{u}{(1 - u^3)^{1/3}} + C.$$

For the second,

$$\int (1-u^3)^{1/3}\,\mathrm{d}u = \int \mathrm{c}\cdot\mathrm{c}^2\,\mathrm{d}x = \int \mathrm{c}^3\,\mathrm{d}x.$$

The product result in (ii) says $\mathrm{c}^3 = \frac{1}{2}\left(1 + \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{sc})\right)$, so

$$\int \mathrm{c}^3\,\mathrm{d}x = \tfrac{1}{2}(x + \mathrm{s}(x)\mathrm{c}(x)) + C.$$

Returning to u, with $x = \mathrm{s}^{-1}(u)$ and $\mathrm{sc} = u(1-u^3)^{1/3}$:

$$\int (1-u^3)^{1/3}\,\mathrm{d}u = \tfrac{1}{2}\left(\mathrm{s}^{-1}(u) + u(1-u^3)^{1/3}\right) + C.$$

Checks

- The answers in (v) can be checked by differentiating with respect to u. For example,

$$\frac{\mathrm{d}}{\mathrm{d}u}\,\frac{u}{(1-u^3)^{1/3}} = \frac{1}{(1-u^3)^{1/3}} + \frac{u^3}{(1-u^3)^{4/3}} = \frac{1}{(1-u^3)^{4/3}}.$$

- Solving the differential equations numerically up to $x = 1.2$ gives $\mathrm{s}(1.2) \approx 0.9411$ and $\mathrm{c}(1.2) \approx 0.5500$, and indeed $0.9411^3 + 0.5500^3 \approx 1$. With $U = \mathrm{s}(1.2)$, numerical integration gives $\int_0^U (1-u^3)^{-2/3}\,\mathrm{d}u \approx 1.2000$, as (iv) predicts, and $\int_0^U (1-u^3)^{1/3}\,\mathrm{d}u \approx 0.8588 = \frac{1}{2}(1.2 + 0.9411\times 0.5500)$, matching (v).
- Numerically, $\int_0^{1.2} \mathrm{s}^2\,\mathrm{d}x \approx 0.4500 = 1 - \mathrm{c}(1.2)$, agreeing with (iii).

Common slips

- Writing $\int \mathrm{s}^2\,\mathrm{d}x = +\mathrm{c} + C$: the minus sign in $\mathrm{c}' = -\mathrm{s}^2$ is easy to lose.
- In (iv) and (v), forgetting to convert $\mathrm{d}u$ into $\mathrm{c}^2\,\mathrm{d}x$, or leaving the final answers in terms of x rather than u.
- Trying to integrate c^3 directly in (v) instead of spotting that (ii) already expresses it as a derivative plus a constant.

Beyond STEP

The pair s, c behaves like sin and cos for the curve $x^3 + y^3 = 1$ in place of the circle $x^2 + y^2 = 1$; such "generalised trigonometric functions" are close relatives of the elliptic functions, which arise by inverting integrals like the one in (iv), just as $\sin^{-1} u = \int (1-u^2)^{-1/2}\,\mathrm{d}u$. Defining functions through a differential equation with initial conditions, and deriving all their properties from it, is a standard first-year technique, and the assumption that the functions are "uniquely defined" is the existence and uniqueness theorem for ODEs. Boyce and DiPrima, *Elementary Differential Equations*, covers both the theory and many examples of coupled systems.

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